

João Pedro da Conceição Simas

Fullstack Engineer — Python · TypeScript · AI Systems

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SUMMARY

Fullstack engineer with 4+ years of hands-on experience and 1 year shipping production systems at scale. Currently sole engineer on a Python/FastAPI ERP — **2,190 commits across 30+ modules in 11 months** — including an LLM-powered WhatsApp agent, facial biometric authentication, and bank statement parsers for 5 Brazilian banks. Strong background in systems design, AI integration and CI/CD automation.

EXPERIENCE

Fullstack Engineer — AV09 Foreign Trade (ERP — Nexus / USFORCE8) *Itajaí, BR · Apr 2025 – Present*

- Built **SOWA**, a production WhatsApp AI agent: deterministic regex bypass before LLM call, normalization of invisible Unicode marks injected by WhatsApp Web, opaque LID resolution for Evolution API v2, 13 admin tools, Whisper transcription and web search.
- Designed statement parsers for **5 Brazilian banks** (BB, Bradesco, Itaú, Santander mTLS+OAuth2, Sicoob) with SHA-256 deduplication, multi-currency support and AI-assisted PDF parsing.
- Implemented **facial biometric auth** using MTCNN + FaceNet/InceptionResnetV1 with 4-criteria simultaneous matching: cosine similarity, Euclidean distance, separation margin and cohort scoring against top-5 similar faces.
- Architected **CI/CD pipeline from scratch**: infra-files-guard workflow, automated PR review, self-hosted runners — reduced deploy time from **35 minutes to 33 seconds**.
- Delivered 30+ backend modules: audit system covering 210+ routes, delivery routing with Valhalla truck engine, gamification ranking, schedulable pg_dump backup, exchange rate service and server metrics history.

Studies & Personal Projects *Remote · Apr 2024 – Mar 2025*

- Deepened AI/ML and software engineering skills — CS50's Introduction to Artificial Intelligence with Python and CS50's Introduction to Programming with Python (HarvardX) — and built open-source personal projects (Cashly, Prozes, among others), publishing a package on PyPI.

Fullstack Engineer (Freelance) *Remote · Jun 2022 – Mar 2024*

- Delivered web, automation and data science projects for clients across multiple industries under NDA.
- Managed systems serving **tens of thousands of active users**; significantly cut critical process times and operational costs through infrastructure optimisation.

SELECTED PROJECTS

Cashly · *Kotlin · Jetpack Compose · Supabase Realtime · TypeScript · React* · cashly.jaopd.dev

Android finance app that captures transactions automatically by reading bank push notifications in real time. Custom notification parser with intelligent categorisation and continuous learning, offline-first sync via WebSocket Realtime, biometric auth and recurring transactions with AlarmManager.

Prozes · *Python CLI · Published on PyPI* · prozes.jaopd.dev

Scaffolds production-ready Flask / FastAPI / Django projects in seconds, with test cases, virtualenv setup and CI/CD workflows baked in. Multilingual; open source.

RescueStick · *C++ · Python · Linux* · rescue-stick.jaopd.dev

Professional bootable USB system (Debian Linux) for IT technicians, with 7 tools in C++ and Python: Windows password reset, disk backup and recovery, hardware diagnostics, boot repair, file recovery and malware scanning. Modular architecture, MIT license and 100% test coverage.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Kotlin, C#, PHP, C/C++, Bash
Frontend	React, React Native, Tailwind CSS, Next.js
Backend	FastAPI, Node.js, Express, ASP.NET Core, Django, Flask
Databases	PostgreSQL, MySQL, MongoDB, Redis, SQL Server, Supabase, Firebase
DevOps	Docker, Linux, Git, GitHub Actions, GitLab CI, AWS, Azure, Cloudflare
AI / ML	OpenAI API, Anthropic API, MTCNN, FaceNet, TensorFlow, OpenCV, Whisper, Scikit-learn

EDUCATION & CERTIFICATIONS

Technologist in Systems Analysis & Development — UNINTER, Brazil	1st of 4 semesters · In progress
- CS50's Introduction to Artificial Intelligence with Python — HarvardX (edX) - CS50's Introduction to Programming with Python — HarvardX (edX) - ASP.NET Core MVC — Entra21	

LANGUAGES

Portuguese (native) · English (C1) · Spanish (B2) · Italian (B1)